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Timber Structures for Automation and Human-Robot Collaboration

ABSTRACT

The use of timber has long been known for sinking the carbon footprint of the construction industry. Industrial robots' application in different industries demonstrated the advantages of automating processes. Advances in engineering have enabled high-performance timber-based materials. Building with these high-tech materials in a at least partially automated way offers the potential to serve higher quality buildings with cost and time optimization compared to conventional construction methods.

This paper aims to cognize the state-of-the-art industrial robots for automating the construction process, grounded in the analysis of diverse researches where automation took place in building timber constructions.

In the end, a proposal is presented for automating and implementing Human-Robot Collaboration on a timber construction process so far done by hand.

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Photo by Stefano Bazzoli

1. Introduction

1.1 Motivation

Despite the already industrialized market of prefabricated houses, the timber construction sector still has a low level of automation, especially during on-site assembly tasks. This creates a fragmented workflow between the off-site prefabrication and on-site installation.

Furthermore, timber is not among the predominant materials in German construction, despite of wood being a renewable raw material and having environmental and climate-policy advantages. New implementations of Mass Timber in construction have augmented the use of this material in non-commercial buildings in the last years. Though, the processes still include high proportions of manual work.

Construction is a labor-intensive sector. Robotics, along with automated systems, have the potential to innovate and optimize the German wood industry, the same way they have done it in other sectors. Their efficiency has lowered labor costs, improved productivity, working environment, and quality.

This paper aims to examinate the industrial robotic market, to identify appropriate machinery, with potential to automate and collaborate with construction workers in both, fabrication and assembly tasks in the timber construction industry through automation and Human-Robot Collaboration (HRC).

1.2 Goal

The purpose of the research is to cognize which industrial robots might aid the automation and human-robot collaboration of the timber construction process on- and off-site in Germany.

2. State of the art

2.1 Building Materials in the Construction Industry

To select and categorize the most suitable industrial robots available, it is necessary to understand the material available in the market. This section gives a picture on the different types of wood one has access in Germany, as well as the situation regarding building materials in the housing market.

When it comes to buildings, Germany's predominant materials are steel, reinforced concrete and masonry [1]. Although wood is not among the most common material used in construction, the quota of timber has reached the 16.1% [2] and varies according to the state (Figure). Being Baden-Württemberg the state with the largest share of timber constructions thanks to the successful implementation of its "Model Building Code" (Musterbauordnung, MBO), first published, executed in 2002 [3].

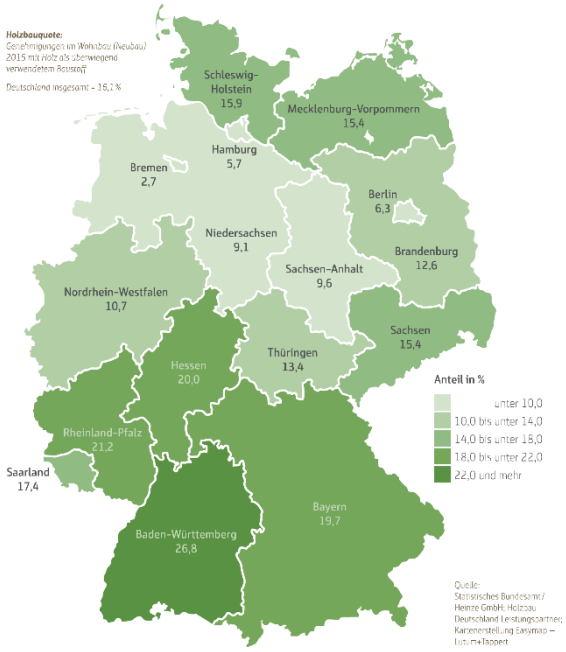

Figure 1: Timber construction quota: Permits in residential construction (new construction) 2015 with wood as the predominantly used building material [5]. Credits: Deutscher Hozlwirtschaftsrat e.V.

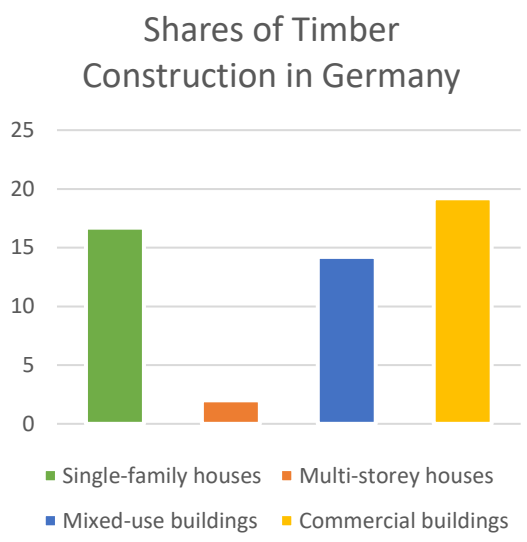

Figure 2: Shares of Timber construction in Germany (2016) [6].

According to the German Timber Council (Deutscher Holzwirtschaftsrat or DHWR), over 35 million m³ of round wood are processed. 20 million m³ of the produced sawn timber were coniferous and 1 million m³ were hardwood. An average of 75-80% of the timber is destinated to the construction industry [2] [4].

From the total of building permits granted in 2016, around 16.7% of single-family houses in Germany were built with timber, only 2% of the non-residential buildings, 14.2% of the mixed-use buildings and 19.2% of commercial buildings [5].

According to the Federal Statistical Office (Destatis) [6], 22.2 percent of the 105,962 newly approved single and two-family houses in Germany in 2020 were prefabricated houses. Unsurprisingly, Baden-Württemberg leads the table with a prefabricated share of 38.4 percent.

2.2 Timber Material in Construction

Wood has been used to build since forever, the first timber house dates back to the Stone Age [7]. With the pass of time and industrialization, wood became to appear as unsafe and weak especially after disasters like the Great Chicago Fire of 1871 [8]. Timber was widely replaced by more modern materials: concrete and steel. They took over the industry thanks to their strength, pliability, permanence, convenience, and relatively cheaper prices [9] [10].

However, in virtue of the advances in engineering in terms of Mass Timber products, new techniques can be not only applied but have the possibility to automate and create high-performance buildings [11]. These efficient high-tech wood-based materials pose a compelling alternative to traditional building materials due to their sufficient load-bearing capacity, sustainability, physical and mechanical properties [12] [13] [14] [15] [16]. Timber-concrete composite (TCC) also has shown potential to develop into a technical solution for the construction sector [1].

As a result, based on the type of obtainable wood in the region, common construction systems in Germany [11] [17], and the available engineered timber products, the timber construction systems and roof structures where industrial robots could play an important role in the residential building sector are listed below:

- Timber Construction Systems:
  - Timber Framing
  - Skeleton Construction
  - Timber Panel Construction
  - Block Panel Construction
- Roof Structures:
  - Rafter
  - Collar Beam
  - Purlin

All these processes can use different connection methods [11] [18]. Regardless of the type, all of them need high precision, a maximum tolerance of about ±0.3mm is required for robots. [19].

While prefabricated timber elements for the systems above are automated because of mass production, mass timber trusses are more complex and thus, still designed and handmade by carpenters and craftsmen.

2.3 Industrial Robots and Cobots

Robotic systems can be divided into three main categories. Industrial Robots fall into of the group of “Robotic manipulation

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systems" and are mostly used in the manufacturing industry [20].

Industrial robots are characterized for their accuracy, flexibility, and speed, bringing great support to any industry that uses them. In terms of construction, they contribute with advantages in productivity, safer work environments, reduction of production and labor cost, as well as increasing the production speed [21]. Although the initial cost of implementation of robots are high, not to mention the need for highly-qualified specialists and their required constant training, costs are still low, compared with the benefits which industrial robots could provide long term [22].

There are multiple options in the Industrial Robots market. For this paper only articulated robots will be taken into account due to their degrees of freedom, versatility when performing different functions and possibility to adapt in the generally unstructured environment of the construction site [23].

Collaborative Robots (Cobots) refer to robots that allow physical interaction with humans, supporting them in repetitive or dangerous tasks. These usually lightweight robots are easy to program, configurate and install [25] and thus, people with little or no expertise in robotics could use them. Robots in human environments need a control framework and diversified interfaces, which must provide enough sensory feedback to the human [26]. These interfaces grant Cobots to learn by human motion tracking, hand guiding control, demonstration (imitation learning), teach pendant, teleoperation and machine to machine interfaces [27]. Nonetheless, because of their limited reachability and handling capacity, Cobots are not as precise or fast and cannot perform as many tasks in the wood construction industry as traditional industrial robots [25].

Figure 3: KUKA Articulated Robot fabricating components for a pavilion at the 2019 Bundesgartenschau [24]. Credits: Katharina Muench / KUKA Robotics

There is a possibility to adapt traditional industrial robots to achieve HRC. This though involves a great economic investment [28]. Nevertheless, previous work shows that, radio control, cameras and GPS sensors installed in construction machinery improved operator safety. A step further would be to use high-level commands such as voice and haptic for intuitive control although, this technology is rarely used in construction [26].

2.4 Automation and Industrial Robots in Timber Construction

Automation and digitalization have benefited the construction industry in the last few years, this however is mostly focused on traditional and standardized mass-production and not so much on product customization or on-site construction processes [29]. The techniques, tasks and processes during assembly stages and the building's lifecycle in general, are yet mostly manually done due to a high flexibility requirement and knowledge that only humans can contribute [30].

In the industry's current state, the main phases in a simple modular building construction and their potentials for an automated construction site are shown in Figure [27]. Subsequently, the processes involving timber structures are as follows:

- Module Fabrication
- Panel Fabrication
- Roof Manufacturing
- Module Installation
- Wall (Panel) Installation

To increase automation, the timber construction industry needs a framework

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with integrative design methodologies. The market of prefabricated houses has improved over the last few years, due to CNC systems, but this only applies to highly controlled environments. In processes such as assembly and installation, a high level of manual work is still key. Although there are on-site tasks structured enough for a robot to perform, many others still need manual labor. 85% of current robotics usage in the wood products industry is on the handling side, while 10% of mechanisms are used for assembly and 5% for machinery [30]. One of the reasons for limiting industrial robots to off-site tasks is that construction sites are challenging for robots. Each construction project and its site are unique, unstructured, and dynamic. In consequence, industrial robots are either limited to off-site tasks or, in order to perform on-site jobs, mounted on mobile platforms but performing single, preprogrammed tasks [26]. Mobile robots in addition, require extra elements to compensate errors caused by unexpected moves of the platform due to irregularities on the floor [29].

To achieve automation and HRC in the construction site, the robot needs to understand and, in a way, feel the environment. Haptic feedback is significant for making teleoperation more intuitive [26].

In order to use traditional industrial articulated robot for HRC in the construction site, additional devices are necessary. Sensors and cameras play a key role allowing the robot to detect objects or humans around the working environment. Because the construction site is a demanding environment, it has to be verified that the the data coming from the sensors is reliable. Therefore LiDAR (laser range finder), together with cameras and computer vision algorithms are used to scan the construction site. A digital twin is created in real time in order to avoid problems and collisions, and to detect unexpected changes in the layout of the site [31]. Some other sensors relevant for these tasks are proximity, contact, distance (laser based), GNSS (GPS), tactile, ultrasonic and electromagnetic [28].

3. Methodology

The methodology of this research is divided in three parts: Firstly, based on case studies of articulated industrial robots used in timber construction projects, the characteristics of robots used in the timber industry are identified. The scrutiny focuses

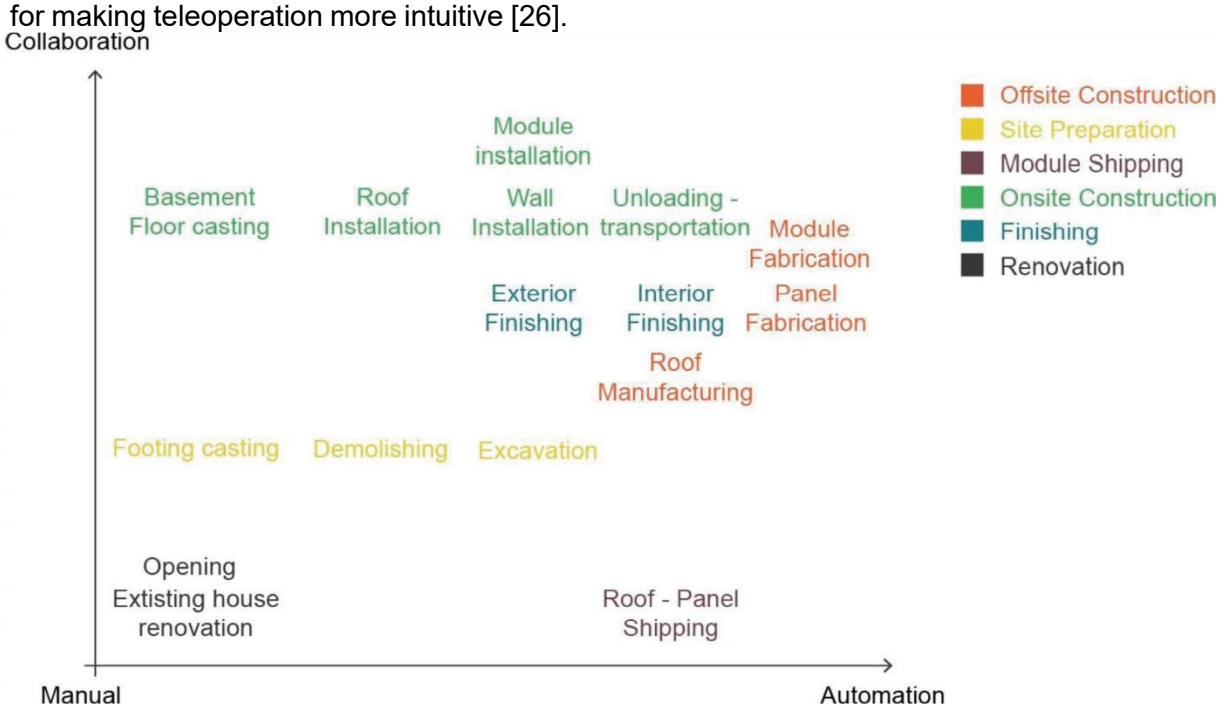

Figure 4: Potentials on automation and collaboration of current construction processes. Credits: Emre Ergin and Kar Yen Lee based on [27]

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Project Code	Project Reference Name	Institute	Year	Element	Span of the Element	Weight of the Element	Robot used	Amount of Robots	Additional equipment for Robots
RT1ETH	*Robotic prefabrication of timber structures*	ETH	2017	Timber slats/ beams	400-1500 mm/ Up to 5000m	N/A	IRB 4600-40/2.55	2	ABB IRBT 2005 - 5m
TA2ETH	*Spatial Timber Assemblies*	ETH	2018	Timber Beams 160x80 mm	Up to 8100 mm	Up to 55kg	IRB 4600-40/2.55	2	+3-axis gantry FP (Güdel)
BU3UOS	*BUGA Wood Pavilion*	Stuttgart	2019	Hollow cassette building assemblies	1500-2700 mm diameter	Up to 200kg	KR FORTEC 500 R2830	2	+TIM
AH4RWT	*Adaptive Haptically Informed Assembly*	RWTH	2019	Timber pieces 35x35mm	N/A	N/A	*LBR iiwa* 14 R820	1	+KMR
AR5UOS	*Implementation of an AR workflow for H-R Collaboration*	Stuttgart	2019	Timber piece	N/A	N/A	*LBR iiwa* 14 R821	1	+KUKA flexFELLOW +Microsoft HoloLens ARHMD
RT6UCA	*Robotic timber assembly*	UCLA	2020	Timber piece 100x100mm	2430 mm	N/A	KR QUANTEC 150-2	2	+6m tracks x2
FT7DTU	*Design Computation to Robotic Production*	DIA/ TU Delf	2020	Timber piece 30mm x 60mm	500-1700 mm	N/A	KR CYBERTECH 16 R1610	1	-
RT8ETH	*Robotic assembly of Timber joints*	ETH	2021	Spruce Slats 40x80mm	1200 mm	1.6kg	IRB 4600-40/2.55	1	-

Table 1: Articulated industrial robots used in timber construction projects.

on product customization. Instead of mass production.

In the second part, a competitive analysis is carried out on the available industrial robots in the market, finding the articulated industrial robots with relevant abilities and KIPs for working with timber. Afterwards, built upon the results from the case studies and the competitive analysis, the six most competent articulated industrial robots, judged by their versatility when working with timber, are determined.

The third section uses radar charts, to conduct a qualitative analysis of the performance of the previously selected most appropriate robotic arms for automating timber construction processes. This is done after adapting the traditional robots for HRC. The determinators will be the construction processes as explained in section 2.5, including the Timber Construction Systems and Structures defined in section 2.3.

Lastly, a specific timber process is analyzed in the discussion, using one of the selected industrial robots in the construction process of a timber roof truss. In addition, the research proposes a concept for implementing HRC in an on-site production process of the truss.

4. Results

4.1 Part 1: Comparison of industrial robots used in timber construction projects.

The scope consists of eight projects done over the last 5 years, where the focal point was the automation of timber constructions, as well as human-robot collaboration. On-site and off-site work was considered. The subsequently selected projects are:

1. Robotic prefabrication of timber structures – *ETH* [32]. (RT1ETH)
2. Spatial Timber Assemblies / DFAB House – *ETH* [33] [34]. (TA2ETH)
3. BUGA Wood Pavilion – *University of Stuttgart* [35] [36]. (BU3UOS)
4. Adaptive Haptically Informed Assembly with Mobile Robots in Unstructured Environments– *RWTH* [29]. (AH4RWT)
5. Implementation of an Augmented Reality AR workflow for Human Robot Collaboration in Timber Prefabrication – *University of Stuttgart* [37]. (AR5UOS)

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ABB Group

No.	Robot Series	Maximum Reach (mm)	Payload (kg)	Weight aprox. (kg)	Number of Axes	Repeteability Accuracy ± (mm)	Cobot
1	*IRB 1100* 4/0.475 - 4/0.58	475-580	4	21	6	0.01	
2	*IRB 120* -3/0.6	580	3	25	6	0.01	
3	*IRB 1200* -7/0.7 - 5/0.9	700-900	7-5	52-54	7	0.02-0.025	
4	*IRB 1300* -11/0.9 - 7/1.4	900-1400	11-7	74.5-78.5	6	0.02-0.03	
5	*IRB 140 - IRB 140T*	810	6(+)	98	6	0.03	
6	*IRB 1600* 6/1.2 - 10/1.45	1200-1450	6-10	250	6(+)	0.02-0.05	
7	*IRB 2400* /10 - /16	1550	10-16	380	6	0.03	
8	*IRB 2600* 20/1.65 - 12/1.85	1650-1850	20-12	272-284	6+3(+)	0.04	
9	*IRB 4400* /60	1960	60	1040	6	0.06	
10	*IRB 4600* -6/2.05 - 20/2.50	2050-2510	60-20	425-412	6+3(+)	0.06-0.05	
11	*IRB 6620*	2200	150	900	6	0.03	
12	*IRB 6650S* -125/3.5	3500	125	2250	6	0.9	
13	*IRB 6660* -100/3.3 - 205/1.9	3350-1930	100-205	1950-1730	6	0.10-0.07	
14	*IRB 6700* -200/2.60 - 245/3.00	2600-3000	200-245	1250-1540	6	0.05	
15	*IRB 6700Inv* -245/2.90	2900	245	1705	6	0.06	
16	*GoFa* CRB 15000	950	5	27	6	0.05	✓
17	*SWIFTY* CRB 1100-4/0.58	580	4	21	6	0.01	✓
18	*YuMi* IRB 14000	559	0.5	38	6	0.02	✓

Table 2: Potential ABB Industrial robot series for Timber construction.

KUKA Robotics

No.	Robot Series	Maximum Reach (mm)	Payload (kg)	Weight aprox. (kg)	Number of Axes	Repeteability Accuracy ± (mm)	Cobot
1	*KR 3 AGILUS* R540	541	3	26.5	6		
2	*KR 6 AGILUS* R900-2	901	6	53	6		
3	*LBR iiwa* 7R800 - 14 r820	800-820	7-14	22.3-29.5	7	0.1	✓
4	*KR CYBERTECH nano* 6 R1840-2 - 10 R1440-2	1840-1440	6-10	162-153	6	0.04	
5	*KR CYBERTECH* 8R2010-2 - 22 R1610-2	2013-1612	8-22	260-255	6	0.04	
6	*KR IONTEC* 20 R3100 - 70 R2100	3101-2101	20-70	549-536	6	0.05	
7	*KR QUANTEC* 120 R2700-2 - 270 R2700-2F	2701-2696	167(+) - 270(+)	1069-1101	6	0.05	
8	*KR QUANTEC* 120 R3900-2K - 210 R3300-2K	3904-3305	146(+) - 273(+)	1280-1266	6	0.05	
9	*KR QUANTEC nano* 120 R1800 - 160 R1570	1803-1573	120(+) - 210(+)	684-677	6	0.06	

Table 3: Potential KUKA Industrial robot series for Timber construction.

6. Robotic timber assembly – *UCLA* [38]. (RT6UCA)
7. Design Computation to Robotic Production Methods for Reciprocal Tessellation of Free-from Timber Structures – *DIA / TU Delf* [39]. (FT7DTU)
8. Robotic assembly of timber joints – *ETH* [40]. (RT8ETH)

In addition, each project was given a code for convenience of this research to address to them in the upcoming analysis.

Table 1 shows the span of the timber elements treated on each project, as well as their weight. The robots implemented in the construction of the respective timber

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SWIFTI CRB 1100-4/0.58	
KPIs	**Suitable Applications**
▪ **Max. Reach:** 580mm ▪ **Payload:** 4kg ▪ **Weight:** 21kg ▪ **No. of Axes:** 6 ▪ **Repeatability Accuracy:** ± 0.01mm ▪ **Velocity:** 5.05m/s ▪ **Sensors:** Yes ▪ **Safety:** Supervised by SafeMode, safety laser scanner, ▪ **Potential for H-R collaboration:** Yes ▪ **On-Site work:** Yes ▪ **Mobile platform:** No	▪ Cobot ▪ Assembly ▪ Material Handling ▪ Milling ▪ Cleaning/Spraying ▪ Screw-driving ▪ Polishing ▪ Grinding ▪ Buffing ▪ Deburring ▪ Sanding

IRB 4600 40/2.55	
KPIs	**Suitable Applications**
▪ **Max. Reach:** 2550mm ▪ **Payload:** 40kg ▪ **Weight:** 435kg ▪ **No. of Axes:** 6 ▪ **Repeatability Accuracy:** ± 0.06mm ▪ **Sensors:** No ▪ **Safety:** Doble circuits with supervision, emergency stops. ▪ **Potential for H-R collaboration:** Yes ▪ **On-Site work:** Yes ▪ **Mobile platform:** Yes	▪ Assembly ▪ Material Handling ▪ Material Removal ▪ Milling ▪ Dispensing ▪ Cleaning/Spraying ▪ Packing ▪ Laser Cutting ▪ Laser Welding

IRB 6700Inv -300/2.60	
KPIs	**Suitable Applications**
▪ **Max. Reach:** 2600mm ▪ **Payload:** 270kg ▪ **Weight:** 1690kg ▪ **No. of Axes:** 6 ▪ **Repeatability Accuracy:** ± 0.05mm ▪ **Sensors:** No ▪ **Safety:** Doble circuits with supervision, emergency stops. ▪ **Potential for H-R collaboration:** Yes ▪ **On-Site work:** No ▪ **Mobile platform:** Yes	▪ Assembly ▪ Material Handling ▪ Material Removal ▪ Milling ▪ Dispensing ▪ Finishing ▪ Pick and Place ▪ Palletizing ▪ 3D Printing ▪ Remote tcp

Figure 5: Selected ABB Articulated Industrial Robots, KPIs and applications.

LBR iiwa14 R820	
KPIs	**Suitable Applications**
▪ **Max. Reach:** 820mm ▪ **Payload:** 14kg ▪ **Weight:** 29.5kg ▪ **No. of Axes:** 7 ▪ **Repeatability Accuracy:** ± 0.01mm ▪ **Velocity:** 5.05m/s ▪ **Sensors:** Yes ▪ **Safety:** Torque sensors: contact detection and automatic speed reduction. ▪ **Potential for H-R collaboration:** Yes ▪ **On-Site work:** Yes ▪ **Mobile platform:** Yes	▪ Cobot ▪ Teaching by demonstration ▪ Applying/Painting/ Glueing ▪ Assembly ▪ Palletizing/Pressing/ Packaging ▪ Material Handling ▪ Milling ▪ Polishing ▪ Mechanical Machining ▪ Polishing

CYBERTECH KR20 R1810-2	
KPIs	**Suitable Applications**
▪ **Max. Reach:** 1813mm ▪ **Payload:** 20kg + 15kg arm suppl. load ▪ **Weight:** 255kg ▪ **No. of Axes:** 6 ▪ **Repeatability Accuracy:** ± 0.04mm ▪ **Sensors:** No ▪ **Safety:** Emergency stop. ▪ **Potential for H-R collaboration:** Yes ▪ **On-Site work:** Yes ▪ **Mobile platform:** Yes	▪ Assembly ▪ Handling large components. ▪ Material Removal ▪ Mechanical Machining ▪ Cutting/Separation ▪ Palletizing/Pressing/ Packaging ▪ Laser welding/ cutting ▪ Applying/Painting/ Glueing ▪ Polishing ▪ Spot/Arc welding

QUANTEC KR270 R2700-2F	
KPIs	**Suitable Applications**
▪ **Max. Reach:** 2696mm ▪ **Payload:** 270kg + 50kg arm suppl. load ▪ **Weight:** 1129kg ▪ **No. of Axes:** 6 ▪ **Repeatability Accuracy:** ± 0.06mm ▪ **Sensors:** No ▪ **Safety:** Emergency stop. ▪ **Potential for H-R collaboration:** No ▪ **On-Site work:** Yes ▪ **Mobile platform:** Yes	▪ Applying/Painting/ Glueing ▪ Assembly ▪ Palletizing/Pressing/ Packaging ▪ Material Handling ▪ Material Removal ▪ Cutting/Separation ▪ Laser welding/ cutting ▪ Mechanical Machining ▪ Polishing ▪ Spot/Arc welding

Figure 6: Selected KUKA Articulated Industrial Robots, KPIs and applications.

structure, together with any extra equipment for the robot listed there. N/A is written on the table since some of the information regarding the timber elements was not given in the researched literature.

It can be deduced from Table that the robots used in these projects, based on their strength fall into the category of Low, Medium, and Heavy payload. The largest and most powerful articulated robot is KR FORTEC 500 R2830, used for the project BU3UOS. However, the heaviest element lifted by the robot weighed around 200kg [19], meaning that the robot worked at 40% of its full capacity thus, a smaller series robot could have been used as well. For

instance, a robot from the QUANTEC series, such as the KR270 could replace it since both have similar KPIs, but the weight and cost of the KR270 is lower, making it more suitable for transportation in case of possible on-site work.

4.2 Part 2: Sample Robots from the Industrial Robot Market.

In order to limit the scope size, the research focuses on two leading companies in the European industrial robot market [41]. They are:

- ABB (Switzerland) [42]
- KUKA (Germany) [43]

The maximum size limit is based on the size of the robotic arms available at RWTH University for the Construction and Robotics program, with the smallest being the KUKA KR3 AGILUS R540, and the biggest the KUKA, KR 270 R2700F. It has to be considered as well that the main applications of the robots are relevant for working with timber and their potential for Human-Robot collaboration.

Table 2 and 3 document the assessed available articulated robots offered by the companies mentioned above.

Derived from the comparative tables and the categories obtained in 4.1 Part 1, the six most versatile articulated robot series to work with timber are shown in Figure 5 and 6.

In pursuance of choosing the sample robots from the selected series, the payload was considered for the Low and High robots. In contrast, for the medium ones, maximum reach was also key factors.

4.3 Part 3: Performance of Industrial Robots in Automation and HRC

In this section, the performance of the articulated robots is analyzed for automation and human-robot collaboration in traditional framing, panel and roof construction.

Taking the scoped projects as examples, the most accurate way to use articulated robots for automating the construction process is by complimenting them with additional equipment such as tracks, 1-3 axis gantries or even mobile robots [44].

Moreover, if the necessary sensors described in the literature reviewed in

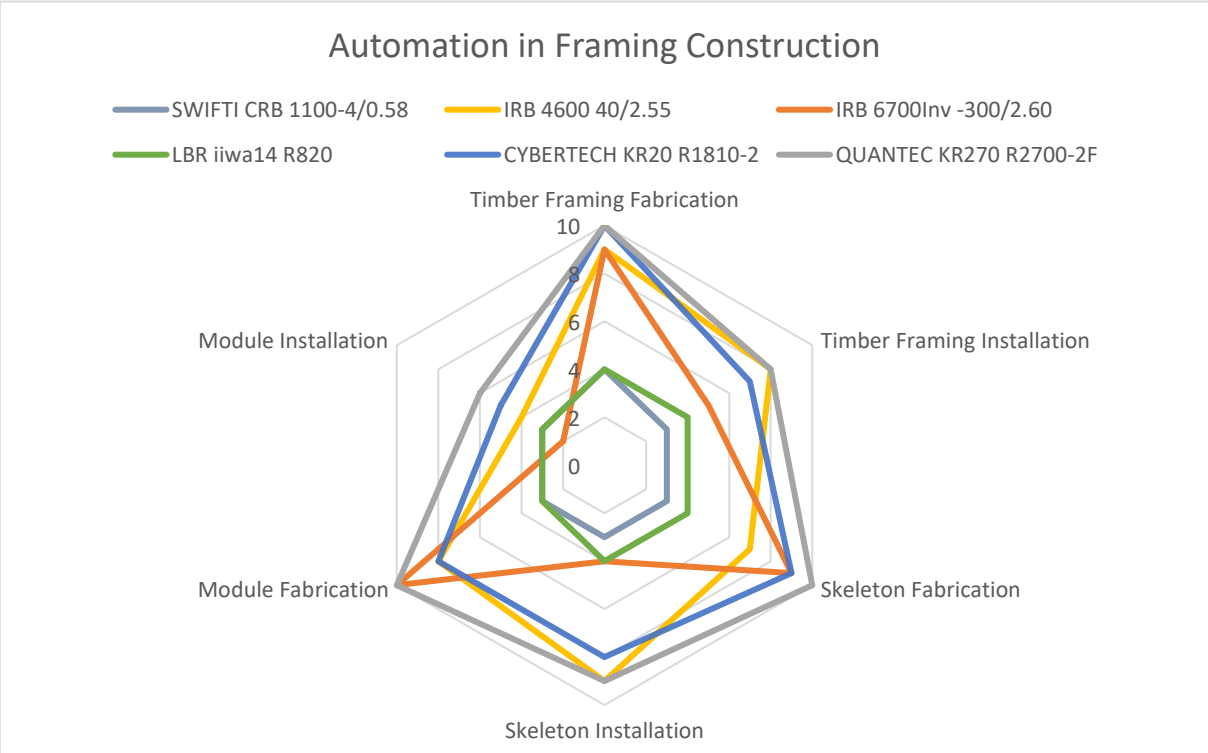

Figure 7: ABB and KUKA robots performing automated tasks in Framing construction

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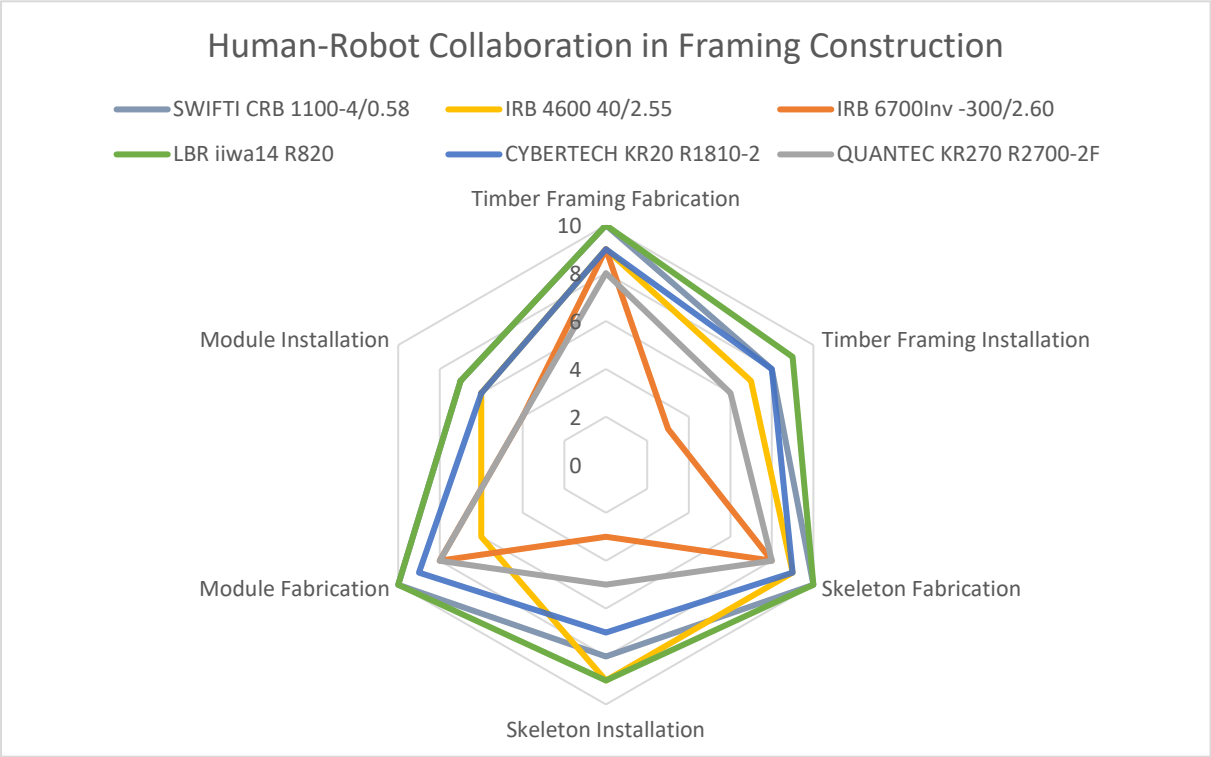

Figure 8: ABB and KUKA robots performing Human-Robot collaboration tasks in Framing construction

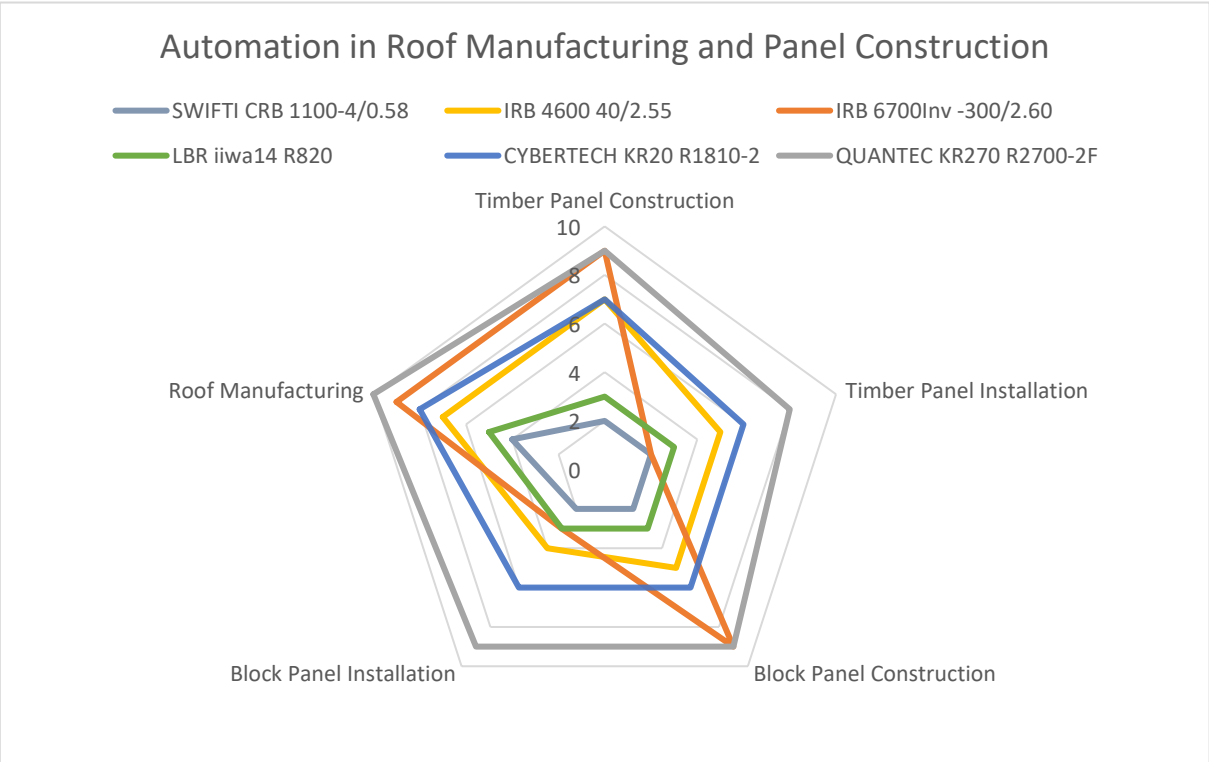

Figure 9: ABB and KUKA robots performing automated tasks in Panel construction and Roof Manufacturing.

chapter 2 are applied, traditional industrial robots could be suitable for HRC in the construction site to a certain degree. Despite of this, because they don't possess integrated sensors, reacting to slight forces and enabling collision protection is not feasible. Therefore, it may not be suitable

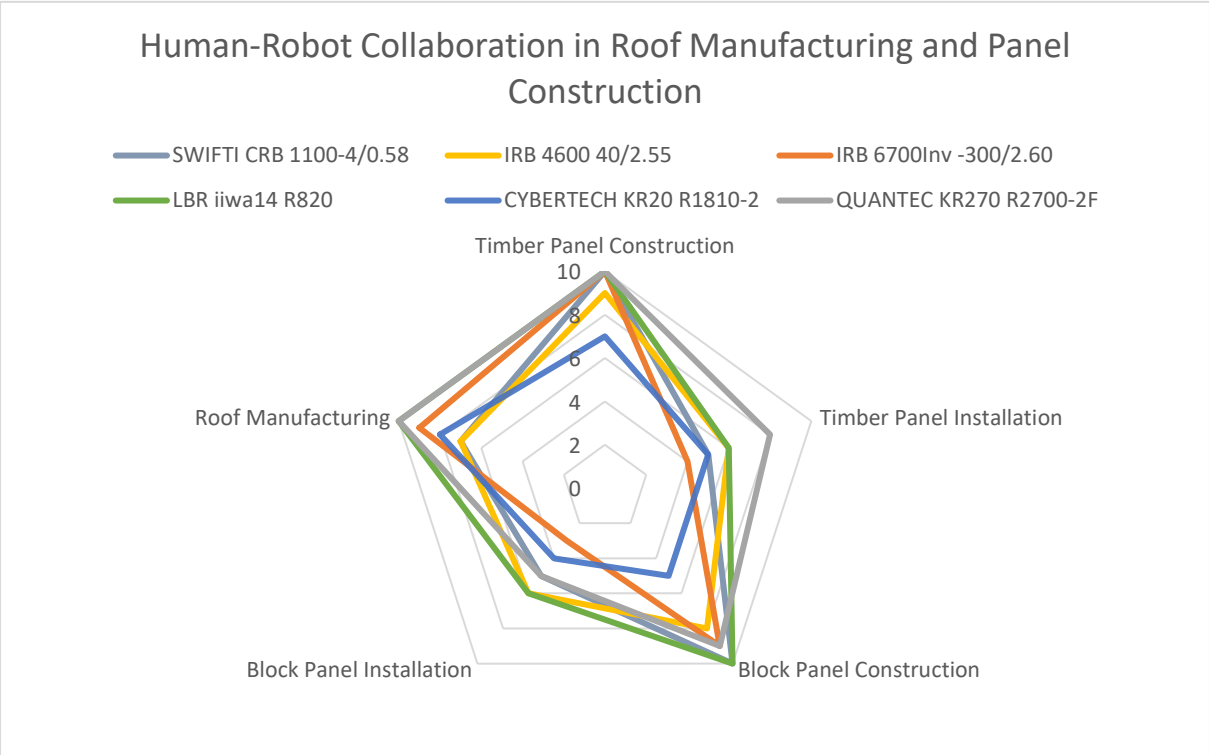

Figure 10: ABB and KUKA robots performing Human-Robot collaboration tasks in Panel construction and Roof Manufacturing.

for certain specific tasks where close interaction with humans is required.

The evaluation for determining the score of each robot in the different fields is influenced by their KPI's. Specifications such as reachability, and payload determines the ability of the robot to work with a wide range of materials and sizes.

Similarly, degrees of freedom, accuracy, speed, and vision and sensors impact the capability of performing tasks with high precision and reaching convenient positions for HRC. Likewise, if compatible tracks and mobile robots are contemplated, the adaptability for on-site non-static tasks also influences the result.

While the previously reviewed projects have a high percentage of automation attributed to their design being influenced by the articulated robot's design and KPI's, traditional construction methods are more difficult to become fully automated. Nevertheless, medium, and especially high payload robots represent a great assistance in fabrication and installation, shown in Figure 7, 8,9 and 10.

On the other side, low payload and in this case, also cobots are not suitable for fully automating the fabrication or installation processes.

In the construction processes described above, both, traditional industrial robots and collaborative robots are able to assist in handling and joining processes. They include precise tasks (positioning, handling, cutting, joinery, milling), as well as repetitive tasks (bolting, drilling, screwing, sanding).

A constant in all the figures in part three is the abilities all robots have for off-site prefabrication, which confirms that traditional timber construction industry follows a fragmented workflow with highly automated processes in fabrication but low when overtaken for installation in situ.

5. Discussion

Researched literature and carried analysis show the achievability to use articulated industrial robots in the timber construction industry. This chapter focuses on a proposal for a workflow concept for the construction of a king post truss (Figure 11). Based in the results of the previous chapter, the QUANTEC KR270 is the most

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suitable for the task. This robot was chosen because of its high accuracy for repetitive tasks where precision is requested, especially for joints manufacturing. Also, its payload and reachability can move the heavy individualized produced parts fit them in place during the assembly process. Another reason is the availability of a mobile robot designed for this KUKA series. Lattermost, its possibility to adapt to an extend for HRC, with the aid of multimodal commands based on cameras and sensors.

The additional equipment needed are a crane and a mobile robot suitable for the KR270 (KUKA omniMove). Instead of a prefabrication off-site of a mass-produced roof, the proposal is the individualized production of an on-site timber roof truss process.

In a traditional workflow for this type of truss (Table), the work is done in the majority by labor force, with the aid of CNC machines and cranes.

This traditional approach has little or no automation due to the customized design. The layout is made by pencil marks on each timber piece, the mortises and notches for joints are cut and cleaned by hand, together with sanding and applying finish to the truss.

The table above shows the tasks where industrial robots can be used (yellow), as well as shipping (green).

The proposal workflow starts with the raw material delivered directly to the construction site. This change alone deletes tasks 8 and 9, saving cost and time wasted in shipping. Then, the workspace is set for manufacture and assembly, considering the range of movement and reachability of the robot, along with suitable power supply and weather protection. As stablished in Chapter 3, tasks 2-6 of Table can be easily performed by a single robot.

Figure 11: Arched King Post Truss [52]. Credits: Vermont Timber Works

The implementation of robots in task 7 and 11 reduces drastically the heavy job since the robot can transport the timber pieces from the manufacturing area to the assembling area. When HRC is needed, the sensors described in chapter 2 are used for internal feedback control and external interaction. For example, the mobile robot may have a LiDAR installed for scanning the site in real life. As for the industrial robot, it could be equipped with camera-based collision prevention, and haptic and voice-based robot control. These adaptations could be used during the truss assembly (mortar and tenon joint): haptic or voice commands are given to re-program

Off-site	1. Raw timber arrives to workshop
	2. Layout
	3. Hand craft / cut timber
	4. Clean timber
	5. Sand Timber
	6. Apply Finnish
	7. Assemble truss
	8. Load trusses for shipment
On-site	9. Delivered timber to construction site
	10. Organize trusses
	11. Assemble bents (if needed)
	12. Lift trusses
	13. Position trusses
	14. Install purlins

Table 4: Traditional workflow process for truss manufacture and assembly

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and re-direct the robot so the tenon aligns correctly with the mortar. Meanwhile, collision detection will prevent the robot to push the handling piece against the rest of the structure and damaging it. With this, manpower and time taken to perform the tasks are reduced. Also, thanks to the accuracy of the robot, the quality and precision of the joints increases significantly.

As mentioned in previous researches [27], it is also crucial to consider capabilities, limitation and status of the workers and team members. A successful HRC is not only based on a powerful robot and latest technology, but also informed employees that relies on the robot's performance, IT security, occupational safety and appropriate allocation of tasks.

6. Conclusion

Timber as a construction material shows promising improvements in ecological, economic and social aspects. The available high-tech wood-based materials with high physical and mechanical properties could be a replacement for concrete and steel elements. Yet in Germany, less than one fifth of the buildings has wood as its predominant material. One of the main reasons is the severe restrictions most states have for timber constructions.

The research showed the feasibility of improving the timber construction industry by implementing articulated robots, not to mention the already existing developments such as CNC systems. It has been demonstrated that timber structures can reach up to fifteen storeys high without complication. With the addition of automated off- and on-site processes in modular structure designs, timber construction could be an innovative force in the German construction industry, given that states adopt a new building code similar to the one implemented in Baden-Württemberg, Switzerland or Austria.

Articulated robots in on-site and off-site construction processes decrease costs, manual labor in repetitive and dangerous tasks, and increase the flexibility to the point not reachable by humans. Though, traditional construction methods are more difficult to become fully automated only by using robots. The main reason is that these methods were planned to be executed by manual labor. The elements were not considered to be manipulated by robots.

However, the possibility to implement Human-Robot collaboration on-site is realistic. Apart from cobots, articulated industrial robots require extra elements like sensors, cameras, and LiDAR installed, creating multimodal communication and thus, allowing a safe and adaptable operation. The integration of these elements, together with tracks and mobile robot manufacturing platforms could also support the location and flexibility of the robots on-site. At the same time, due to the construction environment being unstructured, complex, and dynamic, several factors can influence the industrial robot operation. This requires the implementation of extra hardware and software elements to compensate and mitigate the errors.

There is a disclaimer needed in respect of inverted articulated robots. Despite their considerable KPIs, characteristics and applications, and in spite of their great performance for project TA2ETH, they are only practical for prefabrication involving a workshop environment. As a result, this option might need to be reconsidered, depending on the future direction of the research regarding on-site fabrication.

For the following steps, the suggested concept could be developed, keeping the manufacture and assembly processes as a continuous workflow for individualized timber construction. The characteristics of the elements based on the limitations of a robot instead of a human must be considered. The research could go further, sensors and camera installations on the industrial robot may improve shape and material recognition. This could allow the robot to work with unprocessed timber, where flexibility and adaptability is required.

7. References

[1] M. Knauf, "Market potentials for timber-concrete composites in Germany's

The authors are responsible for the content of the papers.

building construction sector," *Eur. J. Wood Prod.*, vol. 75, 2017, doi: 10.1007/s00107-016-1136-9.

[2] *DHWR | Startseite.* [Online]. Available: https://www.dhwr.de/index.php

[3] SWISS KRONO, *Timber Construction Hindered in Several German States.* [Online]. Available: https://www.swisskrono.de/Artikel/Timber-Construction-Hindered-in-Several-German-States-0911153844.html

[4] Johann Heinrich von Thünen-Institut, *Holzverwendung im Bauwesen: Eine Marktstudie im Rahmen der "Charta für Holz"*. Braunschweig: Johann Heinrich von Thünen-Institut, 2013. [Online]. Available: http://nbn-resolving.de/urn:nbn:de:gbv:253-201312-dn052249-1

[5] Deutsche Holzwirtschaftsrat (DHWR), "Roadmap-Holzwirtschaft 2025: Beitrag der Holzwirtschaft zu einer biobasierten Wirtschaft," 2016.

[6] Statistisches Bundesamt (Destatis), "Bautätigkeit - Fachserie 5 Reihe 1 - 2020,"

[7] *Living with Wood: From the Beginning of Time Infographic - Log Cabin Hub.* [Online]. Available: https://www.logcabinhub.com/living-with-wood-from-the-beginning-of-time/

[8] J. Mayo, *Solid wood: Case studies in mass timber architecture, technology and design*. London, New York: Routledge, 2015.

[9] SWISS KRONO, *Building with SWISS KRONO Engineered Wood Products.* [Online]. Available: https://www.swisskrono.de/Building-Materials/Prefabricated-construction/Building-with-Engineered-Wood-Products-1721223345.html (accessed: 07-Dec-21).

[10] Fizzano Brothers Concrete Products, *New Studies Reveal that Concrete Based Construction is Less Expensive than Wood - Fizzano Brothers Concrete Products.* [Online]. Available: https://fizzano.com/blog/new-studies-reveal-that-concrete-based-construction-is-less-expensive-than-wood/ (accessed: 30-Nov-21).

[11] L. Steiger, *Basics Holzbau,* 3rd ed. Basel: De Gruyter, 2020.

[12] P. Leskinen *et al.,* "From Science to Policy," European Forest Institute, 2018.

[13] P. Da Sousa, Ed., *Structures and Architecture*: CRC Press, 2016.

[14] D. Lacroix, C. Viau, D. Côté, M. Poulin, A. Lopez, and G. Doudak, "Overview on the structural performance of timber structures under the effects of blast loading &#8211 research and design considerations," in *Structures and Architecture*, P. Da Sousa, Ed.: CRC Press, 2016, pp. 67–74.

[15] P. Stapel and J.-W. G. van de Kuilen, "Efficiency of visual strength grading of timber with respect to origin, species, cross section, and grading rules: a critical evaluation of the common standards," *Holzforschung*, vol. 68, no. 2, pp. 203–216, 2014, doi: 10.1515/hf-2013-0042.

[16] V. Wilden and B. Hoffmeister, "Experimental analyses of innovative wood-shear walls under seismic loads," in *IABSE Congress, Christchurch 2021: Resilient technologies for sustainable infrastructure*, Christchurch, New Zealand, 2021, pp. 1222–1229.

[17] T. Brotrück, *Basics Dachkonstruktion*. Basel: De Gruyter, 2017.

[18] K. Zwerger, *Das Holz und seine Verbindungen: Traditionelle Bautechniken in Europa, Japan und China,* 3rd ed. Basel: Birkhäuser/de Gruyter, 2015.

[19] YouTube, *Flower power timber robots: How robots can revolutionize architecture and timber industry.* [Online]. Available: https://www.youtube.com/watch?v=bVVNQ5-3tjw (accessed: 13-Dec-21).

[20] RobotWorx, *Three types of robotic systems.* [Online]. Available: https://www.robots.com/articles/three-types-of-robotic-systems (accessed: 12-Dec-21).

[21] T. Bock and T. Linner, *Construction Robots: Elementary Technologies and Single-Task Construction Robots*. Cambridge: Cambridge University Press, 2016. [Online]. Available: https://

www.cambridge.org/core/books/construction-robots/BCA613ACE148774DA57A94393D11ED99

[22] P. Barosz, G. Gołda, and A. Kampa, “Efficiency Analysis of Manufacturing Line with Industrial Robots and Human Operators,” *Applied Sciences*, vol. 10, no. 8, p. 2862, 2020, doi: 10.3390/app10082862.

[23] International Research Journal of Engineering and Technology (IRJET), “Review-on-Development-of-Industrial-Robotic-Arm,” vol. 04.

[24] Katharina Muench / KUKA Robotics, “Pavilion for BUGA,” [Online]. Available: https://assets.kuka.com/share/238929AE-9471-4BEB-B8B5D015C708A83C/?mediaId=981AABC3-BD49-4D69-A8548B0D281CB57B

[25] K. Afsari, S. Gupta, M. Afkhamiaghda, and Z. Lu, *Applications of Collaborative Industrial Robots in Building Construction*, 2018.

[26] C. Brosque, E. Galbally, O. Khatib, and M. Fischer, “Human-Robot Collaboration in Construction: Opportunities and Challenges,” in *2020 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*, Ankara, Turkey, 2020, pp. 1–8.

[27] E. Ergin and K. Y. Lee, “Potential improvement of on-site construction processes by integrating human-robot collaboration,” 2020.

[28] V. R. Prasath Kumar, M. Balasubramanian, and S. Jagadish Raj, “Robotics in Construction Industry,” *Indian Journal of Science and Technology*, vol. 9, no. 23, 2016, doi: 10.17485/ijst/2016/v9i23/95974.

[29] P. Devadass, S. Stumm, and S. Brell-Cokcan, “Adaptive Haptically Informed Assembly with Mobile Robots in Unstructured Environments,” in *Proceedings of the 36th International Symposium on Automation and Robotics in Construction (ISARC)*, Banff, AB, Canada, 2019.

[30] S.-E. Omer, “Timber Construction and Robots,” in *New Technologies, Development and Application*, Cham, 2019, pp. 163–169.

[31] Prof. Dr.-Ing. Jürgen Rossmann, “Digital Twins for (Construction) Robotics,” Jan. 17 2022.

[32] P. Eversmann, F. Gramazio, and M. Kohler, “Robotic prefabrication of timber structures: towards automated large-scale spatial assembly,” *Constr Robot*, vol. 1, 1-4, pp. 49–60, 2017, doi: 10.1007/s41693-017-0006-2.

[33] A. Thoma, A. Adel, M. Helmreich, T. Wehrle, F. Gramazio, and M. Kohler, “Robotic Fabrication of Bespoke Timber Frame Modules,” in *Robotic Fabrication in Architecture, Art and Design 2018*, J. Willmann, P. Block, M. Hutter, K. Byrne, and T. Schork, Eds., Cham: Springer International Publishing, 2019, pp. 447–458.

[34] *Gramazio Kohler Research.* [Online]. Available: https://gramaziokohler.arch.ethz.ch/web/e/about/aboutInfrastructure.html (accessed: 12-Dec-21).

[35] H. J. Wagner, M. Alvarez, A. Groenewolt, and A. Menges, “Towards digital automation flexibility in large-scale timber construction: integrative robotic prefabrication and co-design of the BUGA Wood Pavilion,” *Constr Robot*, vol. 4, 3-4, pp. 187–204, 2020, doi: 10.1007/s41693-020-00038-5.

[36] H. J. Wagner, M. Alvarez, O. Kyjanek, Z. Bhiri, M. Buck, and A. Menges, “Flexible and transportable robotic timber construction platform – TIM,” *Automation in Construction*, vol. 120, p. 103400, 2020, doi: 10.1016/j.autcon.2020.103400.

[37] O. Kyjanek, B. Al Bahar, L. Vasey, B. Wannemacher, and A. Menges, “Implementation of an Augmented Reality AR Workflow for Human Robot Collaboration in Timber Prefabrication,” in *Proceedings of the 36th International Symposium on Automation and Robotics in Construction (ISARC)*, Banff, AB, Canada, 2019.

[38] J. Koerner-Al-Rawi, K. E. Park, T. K. Phillips, M. Pickoff, and N. Tortorici, “Robotic timber assembly,” *Constr Robot*, vol. 4, 3-4, pp. 175–185, 2020, doi: 10.1007/s41693-020-00045-6.

[39] Mostafavi, Sina, Kastrati, Valmir, Badr, Hossam and Mazlan, Shazwan, *Design Computation to Robotic Production Methods for Reciprocal Tessellation of Free-from Timber Structures: Design, production, and assembly of 100 years Bauhaus wood Pavilion*, 2020. [Online]. Available: http://papers.cumincad.org/cgi-bin/works/Show?ecaade2020_468

[40] A. A. Apolinarska *et al.,* “Robotic assembly of timber joints using reinforcement learning,” *Automation in Construction*, vol. 125, p. 103569, 2021, doi: 10.1016/j.autcon.2021.103569.

[41] Statista, *Global industrial robots - companies by revenue | Statista.* [Online]. Available: https://www.statista.com/statistics/257177/global-industrial-robot-market-share-by-company/ (accessed: 11-Dec-21).

[42] Robotics, *Industrial Robot Supplier and Manufacturer | ABB Robotics.* [Online]. Available: https://new.abb.com/products/robotics

[43] “Industrial Intelligence 4.0_Beyond Automation| KUKA Robotics,” 26-Jun-17, 26-Jun-17. https://www.kuka.com/

[44] “Mobile robots from KUKA,” 06-Mar-17, 06-Mar-17. https://www.kuka.com/en-de/products/mobility/mobile-robots (accessed: 14-Dec-21).

[45] *Timber Frames | Post and Beam Homes | Heavy Timber Trusses.* [Online]. Available: https://www.vermonttimberworks.com/